

YOLOv11n per-class precision / recall / AP@0.5 by input resolution, 5-fold cross-validation

Class (cells: P / R / AP@0.5)	Baseline (640)	Champion (640)	Infer 640 (1280-tr.)	Infer 768 (1280-tr.)	Infer 1024 (1280-tr.)	Champion (1280)
Birdnest	0.88 / 0.88 / 0.92	0.88 / 0.91 / 0.92	0.89 / 0.90 / 0.93	0.91 / 0.91 / 0.94	0.91 / 0.91 / 0.95	0.90 / 0.90 / 0.93
Self-Exploded Insulator	0.82 / 0.53 / 0.63	0.79 / 0.65 / 0.72	0.79 / 0.61 / 0.72	0.85 / 0.65 / 0.74	0.82 / 0.69 / 0.77	0.87 / 0.70 / 0.80
Normal Insulators	0.78 / 0.77 / 0.79	0.80 / 0.78 / 0.81	0.78 / 0.79 / 0.80	0.79 / 0.81 / 0.80	0.83 / 0.82 / 0.83	0.82 / 0.80 / 0.83
Normal Damper	0.76 / 0.57 / 0.66	0.79 / 0.70 / 0.74	0.73 / 0.72 / 0.74	0.78 / 0.72 / 0.76	0.80 / 0.72 / 0.77	0.82 / 0.72 / 0.78
Broken Insulator	0.68 / 0.53 / 0.57	0.81 / 0.56 / 0.60	0.76 / 0.57 / 0.64	0.84 / 0.61 / 0.69	0.88 / 0.66 / 0.74	0.90 / 0.64 / 0.74
Flashover Insulator	0.73 / 0.53 / 0.60	0.75 / 0.60 / 0.67	0.75 / 0.57 / 0.62	0.75 / 0.60 / 0.67	0.77 / 0.65 / 0.71	0.78 / 0.65 / 0.73
Defective Damper	0.64 / 0.45 / 0.50	0.78 / 0.58 / 0.63	0.74 / 0.55 / 0.60	0.77 / 0.65 / 0.70	0.83 / 0.66 / 0.72	0.79 / 0.67 / 0.73
OVERALL (mAP@0.5)	0.76 / 0.61 / 0.67	0.80 / 0.68 / 0.73	0.78 / 0.67 / 0.72	0.81 / 0.71 / 0.76	0.83 / 0.73 / 0.78	0.84 / 0.73 / 0.79

974-image ATLI+eduardos pool, fold means (test folds disjoint; 24-36 Defective Damper test instances per fold). Champion = OBB + all-defect 3× oversample + 15° rotation; Baseline = detection, no oversampling. Infer columns evaluate the 1280-trained Champion at smaller input sizes. Bold: best AP per row.